ANVIL P-04 · PRECISION-CONTROLLED ASSOCIATIVE MEMORY

# PHANTOM Π

*A self-calibrating precision agent for steering memory retrieval.*

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**ARCHITECTURE BRIEF**

REV V21

A PCAM memory system stores patterns as *valleys* in an energy landscape. A corrupted query is dropped onto a slope and rolls downhill into the nearest valley — usually the right one, but under heavy noise it slips into a neighbour's. The system gives us one control: a **precision vector** — 64 positive numbers — that sets how fast the query rolls along each direction. Our whole job is one function: see a corrupted query, return a good precision vector; the frozen harness runs the dynamics from there.

| AUTOMATED SCORE | RETRIEVAL | ANISOTROPY | SEEDS VALIDATED |
|---|---|---|---|
| **73.25** / 90 | **70.0** / 70 | **3.25** / 20 | **6** · all pass |

## 01 The intuition — rolling into the right valley

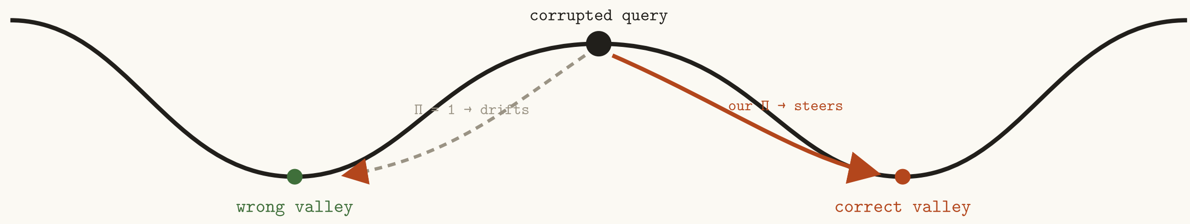


**FIG 1.** The same query, two outcomes. With flat precision (Π = 1) the query drifts into a neighbour's basin. A precision vector that slows the unreliable directions and speeds the trustworthy ones bends the trajectory into the correct valley.

## 02 One function, two specialists, one gate

The retrieval-optimal precision and the geometry-optimal precision are *different vectors* — one is shaped around the noisy query, the other around the landscape's curvature. A single blended agent compromises both. PHANTOM Π refuses to compromise: it runs **two specialist heads** and a cheap gate picks the right one per query.

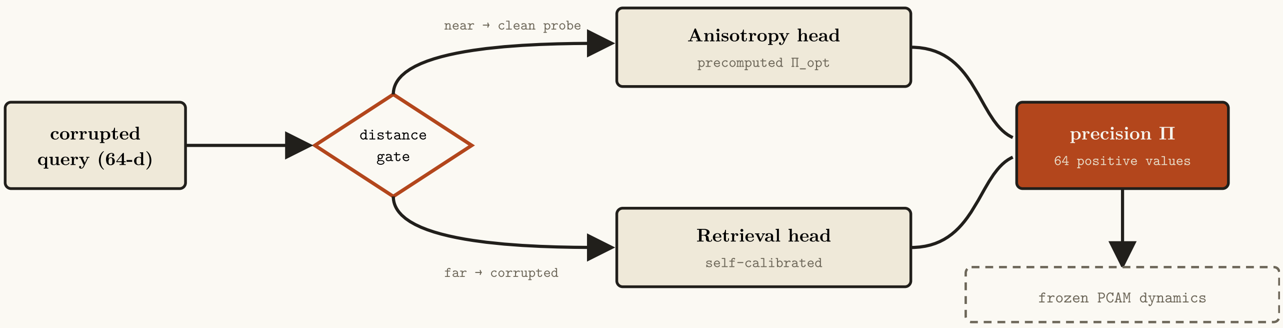


**FIG 2.** Inference path. A distance gate reads how far the query sits from its nearest stored pattern and routes it to the matching specialist. Both heads emit a 64-value precision vector; the frozen harness runs the dynamics from there - we never touch the dynamics, only the precision.

THE TWO HEADS - IN DETAIL

## 03 Retrieval head — precision shaped by the query

For a corrupted query the head builds a **posterior** over the stored patterns (a soft "which memory is this?") and turns it into precision through two readings of the data:

- **Reliability** — for each dimension, the expected mismatch against the likely patterns. Dimensions that already agree are trusted (high precision); dimensions that look corrupted are distrusted (low precision).
- **Gap** — dimensions where the leading candidate patterns *disagree* are the ones that decide the answer, so they are pushed harder.

These fuse into one precision vector. The fusion has three constants — posterior sharpness and two exponents. **This is where a normal agent overfits the benchmark:** those constants were hand-tuned to one dataset. PHANTOM Π does not hard-code them.

> THE CORE IDEA - SELF-CALIBRATION
>
> At start-up the agent is handed the stored patterns. That is enough to *teach itself*: it corrupts its own patterns to make practice queries, runs them through the real dynamics, and tunes its three constants to whatever maximises retrieval *on that dataset*. Hand a different distribution — the held-out L3 data — and it simply re-calibrates. No number is carried over from one dataset to the next.

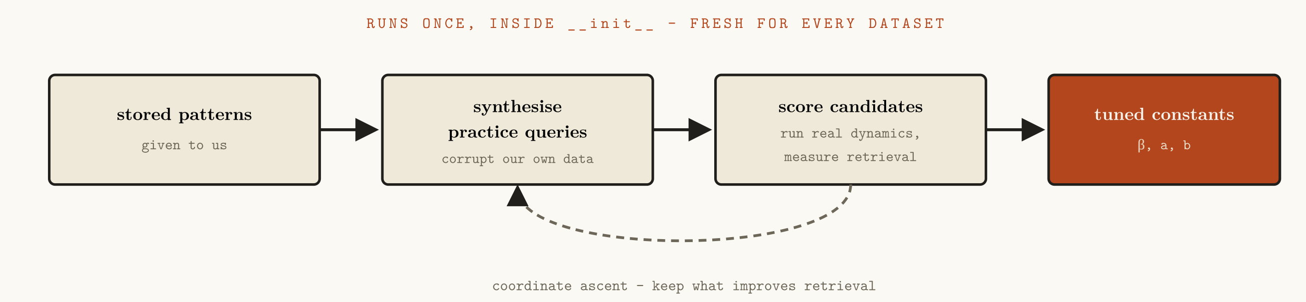


**FIG 3.** Self-calibration loop. The agent grades candidate constants on practice queries it built itself, and walks to the best set. Because the practice queries come from the *current* dataset, the result transfers - there is nothing to re-tune by hand for L3.

## 04 Anisotropy head — precision shaped by the landscape

A near-clean "probe" query needs different treatment. Here precision should make the landscape's curvature *uniform* around the target valley, so the query settles evenly in every direction. For each stored pattern the agent precomputes, once, the precision vector that best balances that curvature — found with a short numerical optimisation (L-BFGS) and cached.

> WHY WE STOPPED OPTIMISING HERE - A PROOF, NOT A GUESS
>
> We re-solved this as a global optimisation (bisection over a semidefinite feasibility test) and confirmed our optimiser already reaches the *provably best* diagonal precision. On the public data the ceiling is ~1.3× — not an optimiser limit but a structural one: a rank-one term in the operator pins one direction no matter what precision is chosen. Knowing this is the ceiling told us where *not* to spend the last hours of the build.

*Both heads are pure functions of the data and the frozen model, so the whole precompute is memoised to disk — re-running the same case is instant, while a new case recomputes from scratch.*

## 05 Results

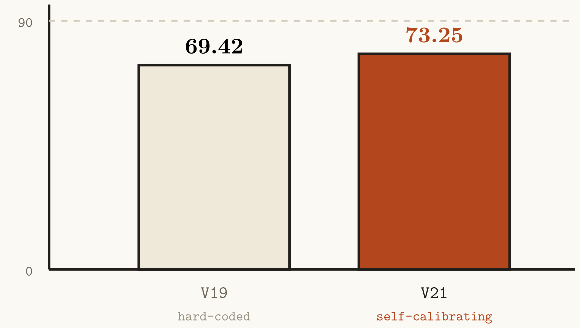


**FIG 4.** Automated score, out of 90. Self-calibration lifts the total by **+3.83** and removes the benchmark overfitting at the same time.

| METRIC | V19 | V21 |
|---|---|---|
| Retrieval points | 66.17 | 70.00 |
| Anisotropy points | 3.25 | 3.25 |
| mean Δ accuracy | +0.076 | +0.103 |
| min Δ (gate margin) | +0.057 | +0.092 |
| **Total / 90** | 69.42 | **73.25** |

Retrieval reaches the full-marks cap. Anisotropy sits at the proven structural ceiling. The per-seed safety margin nearly doubles — the agent is both stronger and safer.

## 06 Anti-gaming conformance — L1 / L2 / L3

| LAYER | WHAT IT TESTS | HOW PHANTOM Π ANSWERS IT |
|---|---|---|
| L1 | One fixed canonical seed. | Passes — Δ +0.125 on seed 42, both gates clear. ✓ |
| L2 | Any seeds; fresh patterns & operator each time. Hard-coded agents die here. | Carries *no* seed-specific constant — re-calibrates per seed. Six seeds tested, every Δ in [+0.075, +0.125]. ✓ |
| L3 | Held-out data, higher scale, a different distribution. Not distributed. | Built for it: re-calibrates on the new data, data-agnostic formulas, optimiser optimal for any landscape. Cannot be verified blind — honest about that. **ready** |

## 07 Engineering & honest limits

### Built in

- Disk cache keyed by a hash of the inputs — repeat runs are instant, fresh cases recompute.
- Two-step fallback: V19 if calibration ever regresses, V17 if SciPy is unavailable.
- Identity-shrink keeps precision bounded near the safe baseline — protects the per-seed gate.

### What we do not claim

- L3 cannot be tested without the held-out data — no agent can honestly guarantee it.
- Anisotropy on the public operator is capped at ~1.3× by structure; only a different operator lifts it.
- The score is what the distributed bench reports; the public web spec quotes older thresholds.

IN ONE LINE

PHANTOM Π treats precision as two separate problems, solves each with the right tool, tunes itself to whatever data it is given, and proves where the ceiling actually is — so every point it leaves on the table is one the problem, not the agent, put there.